

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (4 points) Find the inverse Laplace transform of $\frac{1}{(s+3)^2}$ using the translation formula $\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$ where $f(t) = \mathcal{L}^{-1}\{F(s)\}$.

Solution:
$$\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^2}\right\} = e^{-3t} \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} \\ = te^{-3t}$$

Problem 2. (6 points) Solve the initial value problem

$$x'' - 9x = 4t; \quad x(0) = x'(0) = 0.$$

Solution: Let $\mathcal{L}\{x(t)\} = X(s)$, then
$$\mathcal{L}\{x''(t)\} = s^2 X(s)$$

Thus,

$$\mathcal{L}\{x'' - 9x\} = \mathcal{L}\{4t\}$$

$$s^2 X(s) - 9X(s) = \frac{4}{s^2}$$

$$X(s) = \frac{4}{s^2(s^2-9)}$$

$$\frac{4}{s^2(s^2-9)} = \frac{A}{s^2} + \frac{B}{s^2-9} = \frac{A(s^2-9) + Bs^2}{s^2(s^2-9)} = \frac{(A+B)s^2 - 9A}{s^2(s^2-9)}$$

$$\Rightarrow \begin{aligned} A+B &= 0 \\ -9A &= 4 \end{aligned} \Rightarrow \begin{aligned} A &= -4/9 \\ B &= 4/9 \end{aligned}$$

$$\begin{aligned} \Rightarrow x(t) &= \mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{4}{s^2(s^2-9)}\right\} \\ &= \mathcal{L}^{-1}\left\{-\frac{4/9}{s^2}\right\} + \mathcal{L}^{-1}\left\{\frac{4/9}{s^2-9}\right\} \\ &= -\frac{4}{9}t + \frac{4}{27} \sinh 3t \end{aligned}$$